Gregor Bučar

Senior project engineer in software and automation for GMP pharmaceutical manufacturing

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SUMMARY

I build software in C# and .NET, from desktop apps and web front ends to the services and APIs behind them. Most of what I've built runs in pharmaceutical plants, where the rules are strict and everything has to be validated and traceable. The recipe editor I wrote most of is used at hundreds of installations. My software usually talks to real machines, so I also write PLC code when a project needs it, along with the specifications and test protocols a regulated plant requires.

HOW I WORK

- I love work where the progress is physical, like an arm that moves, a washer that runs its cycle or a building that reports its own climate.
- I don't trust code until it has run against the real thing. On the recipe engine the tests drive a real PLC, and each one is traced to a requirement in the functional specification.
- New things don't scare me. In 2026 alone I rebuilt our recipe engine on .NET 10, set up my own servers behind Cloudflare and learned Siemens TIA Portal for a customer project.
- I'm available for support, late hours included. From October 2026 I'm also on Trac's support team, under mentorship.
- I mentor two students, train customers' operators on site and give internal presentations. I'm still new at mentoring and would like to run more trainings and workshops.

SKILLS AT A GLANCE

C# and .NET	●●●● **Expert, since 2018**	WPF, WinUI 3, ASP.NET Core, EF Core, SignalR
Desktop apps with MVVM	●●●● **Expert, since 2020**	a process database application with a recipe editor, used at hundreds of plants
TypeScript and Node.js	●●●○ **Advanced, since 2022**	Fastify services between the PLC and the browser
Vue and Nuxt	●●●○ **Advanced, since 2023**	operator apps and planning tools, with Tailwind
Automated testing	●●●○ **Advanced, since 2022**	xUnit, Vitest, hardware-in-the-loop against a real PLC
Deployment and hosting	●●●○ **Advanced, since 2024**	Docker, GitHub Actions, Cloudflare, Linux and Windows services
Computer vision	●●●○ **Advanced, since 2023**	training YOLO models, OpenCV, Python
Building with coding agents	●●●○ **Advanced, since 2025**	Claude Code, with a spec and a plan first, tests and my review
PLC programming	●●●○ **Advanced, since 2024**	object-oriented Structured Text in CODESYS, SCL in TIA Portal
ISA-88 batch control	●●●○ **Advanced, since 2020**	PLC phases and the recipe software above them
OPC UA and fieldbuses	●●●○ **Advanced, since 2022**	PROFINET, Modbus RTU and TCP, RS-232, RS-485, EtherNet/IP
GxP validation	●●●○ **Advanced, since 2022**	URS, FS, SDS, OQ, installation qualification, audit trails
Home automation	●●●○ **Advanced, 2026**	Home Assistant APIs, a custom integration, voice, WLED

Expert means I have built and shipped production systems with it for years, **Advanced** means I have shipped production work with it on my own, **Working** means I have used it in real projects, and **Basics** means I have set it up or used it a few times.

EXPERIENCE

Senior Project Engineer, *Trac d.o.o., Šentjernej* — Jun 2024 – present

I worked at Trac as a student from 2018 to May 2024. Trac builds automation, batch-management software and lab robots for pharmaceutical plants.

Batch-manufacturing platform rewrite

2026, WITH CODING AGENTS

What it does. The platform pharmaceutical plants use to write, approve and run batch recipes, with a full audit trail, rebuilt from the ground up in .NET 10.

My part. I'm the top contributor. I built the recipe engine that runs the batches, carried my chart editor over as a shared control, wrote tests against a real PLC, moved the lab robot's services onto the platform and wrote validation documents that got approved.

How. Each recipe step has its own state machine, so Hold, Stop and Abort behave the same everywhere. Run state is saved safely, so after a crash every running batch recovers. Failed API calls are queued and replayed, and about 40 services sit behind interfaces, tested with small fakes. Every change goes spec first, then agents, then my review.

Controllers for the lab sample-preparation robot

2024–26, BY HAND

What it does. The PLC side of the robot. It covers every axis, valve, gripper and instrument on seven CODESYS controllers running on Raspberry Pi.

My part. I wrote almost all of the module programs on the company's ISA-88 library. They drive Mitsubishi MR-J4 servos over PROFINET (CiA-402), SMC step motors over EtherNet/IP, a DH-Robotics gripper over Modbus RTU, and an A&D balance over RS-232 and a chiller over RS-485 through my own object-oriented serial library.

How. 35 operations extend one abstract base, so they all answer the batch host the same way. Serial devices plug in their own parser through an interface, and the interlocks live with the arm. A simulated plant on two Pis tests sequences without hardware. The PLC code is exported to text and reviewed in git, and moving the controllers to a PREEMPT_RT kernel with isolated cores cut PROFINET worst-case latency from 301 to 94 µs.

PLC programs in Siemens TIA Portal

2026, BY HAND, AI-CHECKED

What it does. Two Siemens projects, both in SCL on S7-1500 controllers.

My part. For an IBC washer I wrote seven of the eight cleaning operations on the company's ISA-88 template, with my own code generators and a checker that catches missing tags before compiling. On a clean utilities plant I moved the project to a new electronic-records bridge and rewrote its events so operators get clear messages.

Building management for the company's office building

2026, BY HAND, AI-CHECKED

What it does. A Raspberry Pi running CODESYS controls ventilation, fan coils, water loops, lighting and WC relays, over 9 PROFINET stations, 68 Modbus RTU devices and 9 Modbus TCP pumps, sharing 861 tags with SCADA and Home Assistant.

My part. On top of a colleague's baseline I added 12 climate sensors with warning and alarm bands, statistics per floor, a health check on every serial device and Home Assistant lighting for floors 0 to 4, and did the commissioning fixes.

How. One acknowledge restarts only the failed devices, and failed sensors are left out of the averages. The statistics function is generic and works on arrays of any length. Alarms and outputs are held for 15 seconds after a restart, so a slow link can't trip a pump.

Recipe editor for batch manufacturing

2020–25, BY HAND

What it does. Process engineers draw a production recipe as a sequential function chart, and the plant runs exactly that chart. It runs at hundreds of installations with thousands of users.

My part. I wrote most of the chart editor (step types, branches, connections, auto-arrange and in-place editors that check parameters as you type) and the condition builder with nested AND/ANY rules.

How. The editor lives in its own library behind an interface, and every node builds on a base class with drawing hooks. Conditions are a tree stored as JSON. Every change writes old and new audit records inside the save transaction. Pan and zoom come from an open-source canvas.

Software for the lab sample-preparation robot

2023–26, BY HAND

What it does. A modular robot that prepares pharmaceutical samples by dosing, weighing, homogenizing and extraction. Each module has its own computer and touchscreen.

My part. Six services on one shared core, operator screens in four languages, the audit trail, the arm's service screens with a signed calibration wizard, game-controller arm control, the YOLO tube-detection service, a fleet deploy tool and central logging (Loki, Grafana).

How. Modules plug their own operations into the core (inversion of control), and every command checks itself before writing to the PLC. Each tag has one OPC UA subscription that every screen shares. Modules describe themselves, so the host configures them without code changes.

Installation-qualification tool

2022–26, BY HAND

What it does. Drives PLC outputs and watches inputs live over OPC UA, records who tested each signal and the result, and prints the qualification reports. Used on five large projects.

My part. I wrote about 90% of the desktop app and three quarters of the API, moved the app from .NET MAUI to WinUI 3 and moved both parts to .NET 8.

How. The app uses MVVM with source generators, one base class per device kind and OPC UA behind an adapter. The REST API is layered and versioned, with dependency injection. Temporal tables keep every earlier attempt, and the app warns before any manual PLC write while an interlock is active.

Workload planner 2023–26, BY HAND

What it does. Plans every project task against each person's hours, holidays and days off six months ahead. Replaced a paid planning tool for 29 people, saving about €5,200 a year.

My part. The scheduling logic, the workload views and a cached .NET proxy to Teamwork.

How. It schedules day by day against real capacity, fixed dates first. Unit tests cover the dependency order, and reusable behaviour lives in Vue composables.

Smaller work at Trac 2018–22, BY HAND

What it does. In 2022 I built a project-office dashboard in Angular with Teamwork sign-in and a Node proxy, and a Blazor WebAssembly kiosk that let party guests start a drink recipe on the robot's batch engine over SignalR. From 2018 to 2021 I wrote my first production code in Windows Forms and WPF, along with my first unit, integration and UI tests.

Full-stack developer, student work, *Špica International* Sep 2022 – May 2023

- On a time-tracking SaaS I moved whole screens from AngularJS to Angular with NgRx and built ASP.NET Core endpoints with xUnit tests for the onboarding, pricing and trial screens.

EDUCATION

BSc in Computer and Information Science, Faculty of Computer and Information Science, University of Ljubljana 2019 – 2023

My thesis, *Design and implementation of a decentralised digital building logbook* (2023, mentor prof. dr. Vlado Stankovski), was graded 10 out of 10. It is a web app that keeps facts about building products, contractors and buildings on the OriginTrail knowledge graph, so anyone can check where they came from.

Technical gymnasium (electrical engineering and computing), Šolski center Novo mesto 2015 – 2019

3rd prize and silver award, national ACM computing competition for secondary schools (RTK), 2018, where I placed 6th of 99 in the first group.

OWN PROJECTS

- **Time-tracking and invoicing SaaS** (2026, with coding agents). It has a multi-tenant .NET 10 API with clean architecture, a Nuxt web app, an offline-first Expo mobile app, a tray timer and a browser extension. Tests fail the build if customers' data could mix, and it's deployed with Docker, GitHub Actions and Cloudflare. Free to use.
- **WebSCADA, now EVA** (2024 to 2026, team of five). I wrote most of the gateway to the field equipment (protocol plugins, keeps watching when the server is down), the licensing, a rule engine and the floor-plan editor. It runs at several sites.
- **Home automation** (2026). I run Home Assistant on a Raspberry Pi with my own voice assistant, a custom integration, WLED and local Tuya devices.
- **Booking platform** (2026, with a small team and coding agents). My part is the booking widget a business embeds in its own website (isolated in Shadow DOM, takes on the host's colours and fonts, falls back to a plain link) and add-to-calendar for Google, Apple, Outlook and Microsoft 365.
- **Pub league app** (2026, a fun side project). It runs leagues, tournaments and Elo rankings on .NET and Expo, with push notifications, live updates and automatic deploys to a Raspberry Pi.

LANGUAGES AND MORE

Slovenian (native), English (C2), French (A2 to B1). Driving licence B.

ALL SKILLS

Languages C# (expert, since 2018), TypeScript and JavaScript (advanced, since 2022), Structured Text (advanced, since 2024), SCL (advanced, 2026), Python (working, since 2023), SQL (working, since 2022), C++ (basics, since 2025).

.NET WPF (expert, since 2020), WinUI 3 (advanced, since 2023), ASP.NET Core (advanced, since 2022), Entity Framework Core (advanced, since 2022), SignalR (advanced, since 2024), Background and Windows services (advanced, since 2024), Stateless state machines (advanced, since 2025), .NET upgrades (advanced, since 2024), .NET MAUI (working, 2022 to 2023), Blazor WebAssembly (working, 2022), MediatR, FluentValidation (working, 2026), WinForms (working, 2018 to 2019).

Web and apps Diagram and canvas editors (expert, since 2020), Vue 3 (advanced, since 2023), Nuxt (advanced, since 2024), Tailwind CSS (advanced, since 2023), Angular, NgRx, RxJS (working, 2022 to 2023), AngularJS (working, 2022 to 2023), React Native and Expo (working, 2026), Electron, browser extensions (working, 2026), Translations (working, since 2024), React (basics, 2026).

Architecture REST API design (advanced, since 2022), Node.js with Fastify (advanced, since 2023), A service per machine module (advanced, since 2023), Real time (advanced, since 2023), Offline-first and crash recovery (advanced, since 2023), Plugins loaded at runtime (advanced, since 2024), Event-driven rule engines (advanced, since 2024), Auth and API security (advanced, since 2022), Clean architecture, CQRS (working, 2026), Multi-tenancy (working, 2026), Third-party APIs (working, since 2022).

Design patterns MVVM (expert, since 2020), Dependency injection, inversion of control (advanced, since 2020), State machines (advanced, since 2020), Strategy, command, factory, observer (advanced, since 2020), Repository, pipeline, options (advanced, since 2022), MVC (API controllers) (advanced, since 2022), Template method, adapter, composite, unit of work (working, since 2020).

Data SQL Server (advanced, since 2022), PostgreSQL (advanced, 2026), Temporal tables, stored procedures (working, since 2022), SQLite (working, 2026), MongoDB (basics, 2023).

Industrial automation CODESYS (advanced, since 2024), Object-oriented IEC 61131-3 (advanced, since 2024), ISA-88 (advanced, since 2020), OPC UA clients (advanced, since 2022), PROFINET (advanced, since 2024), Modbus RTU (advanced, since 2024), Serial communication library (advanced, since 2025), RS-232 (advanced, since 2025), Alarms and events (advanced, since 2024), PLC simulation (advanced, since 2024), Manual arm control (advanced, since 2025), Service screens (advanced, since 2025), Operator screens (advanced, since 2020), OPC UA from the PLC side (working, since 2024), RS-485 (advanced, since 2025), EtherNet/IP (working, since 2025), Modbus TCP (working, since 2024), CiA-402 drive profile (working, since 2024), TIA Portal, S7-1500 (working, 2026), PID control (working, 2026), Real-time Linux (working, 2026), Edge hardware (working, since 2023), Building automation (working, 2026), IO-Link (basics, 2026), WinCC Unified (basics, 2026).

Regulated work (GxP) GMP software (advanced, since 2020), Audit trails (advanced, since 2023), Installation qualification (advanced, since 2022), URS, FS, SDS, HDS, user manuals (advanced, 2026), OQ protocols, requirement tracing (advanced, 2026), E-signatures, Part 11, ALCOA+, GAMP 5 (working, since 2023).

Testing xUnit (advanced, since 2022), Vitest (advanced, since 2024), Hardware-in-the-loop (advanced, 2026), Integration tests (advanced, since 2024), Playwright (working, since 2025), Jest, Jasmine (working, since 2022), NSubstitute, architecture tests, Roslyn analyzers (working, since 2024).

Deployment and operations Docker and Compose (advanced, since 2022), GitHub Actions (advanced, since 2025), Automatic deploys (advanced, 2026), Linux servers on Raspberry Pi (advanced, since 2025), Fleet deploys over SSH (advanced, since 2024), Windows services at customer sites (advanced, since 2024), GitHub Pages with my own domains (advanced, since 2025), Cloudflare Tunnel (advanced, 2026), Transactional e-mail (advanced, 2026), Cloudflare Access, DNS, domains (working, since 2025), IIS (working, 2026), MSIX packaging (working, since 2023), Offline installs (working, 2026), Bitbucket Pipelines (working, 2026), Azure Key Vault (working, 2026), Logs and monitoring (working, 2026), Tailscale, nftables (working, 2026), SMS over a GSM modem (working, 2026), Encrypted backups (working, 2026), Mobile builds (working, 2026), Cloud servers (basics, 2026), Cloudflare Workers, Turnstile (basics, 2026).

Home automation Home Assistant APIs (advanced, 2026), A custom Home Assistant integration (advanced, 2026), Voice (advanced, 2026), WLED (advanced, 2026), Home Assistant setup (working, 2026), Tuya plugs, kept local (working, 2026), MCP servers (working, 2026), PLC to Home Assistant (working, 2026), ESPHome (basics, 2026).

AI and vision YOLO training (advanced, since 2023), Building with coding agents (advanced, since 2025), OpenCV (working, since 2025), Datasets and labelling (working, since 2023), LLM integration (working, 2026).

Working with people Technical writing (advanced, since 2023), Git (advanced, since 2020), Training customers (working), Mentoring (basics).

A short version of this CV and the portfolio are at realgregor.github.io.